

The antiderivative and the indefinite integral

Differentiation run backwards — and the constant that makes the answer a whole family of curves.

LESSON 37–40

4 × 40 MIN

ALGEBRA 11, §2.3

P1 · 10.1

LESSON CLOCK

16

24'

26'

30'

46'

38'

The definition	16 min
Why the constant	24 min
Notation	26 min
Particular antiderivatives	30 min
Practice	46 min
Homework and consolidation	38 min

LEARNING OBJECTIVES

- State the definition of an antiderivative and verify one by differentiating.
- Explain why the antiderivative is determined only up to a constant.
- Write the indefinite integral with its constant of integration.
- Find the particular antiderivative through a given point.

TEXTBOOK

National	pp. 173–192
Cambridge	Pure Mathematics 1, pp. 196–202
Workbook	P1 Exercise 10A

01

Explanation

The definition

ANTIDERIVATIVE

F is an **antiderivative** of f on an interval when $F'(x) = f(x)$ for every x in it.

So the question “what is the antiderivative of $2x$?” asks: what did I differentiate to get $2x$? The answer x^2 is checked by differentiating it back.

$F(X)$	AN ANTIDERIVATIVE	CHECK
$2x$	x^2	$(x^2)' = 2x \checkmark$
$3x^2$	x^3	$(x^3)' = 3x^2 \checkmark$
1	x	$(x)' = 1 \checkmark$
0	7	$(7)' = 0 \checkmark$

ALWAYS CHECK BY DIFFERENTIATING

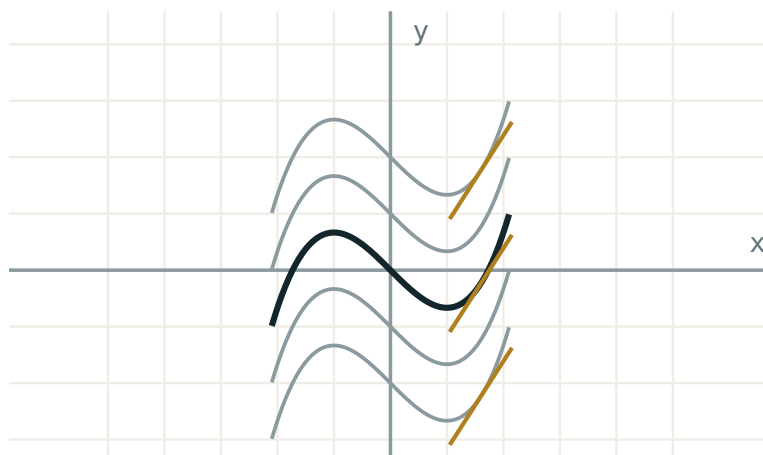
Integration has no algorithm that always works, but its answer can always be verified in one line. That check costs five seconds and catches almost every error.

Why there is a constant

The last row is the clue: 7 is an antiderivative of 0 , and so is -3 , and so is any constant. So if F works, so does $F + C$ — differentiating kills the constant.

$$(F + C)' = F' + 0 = f$$

Theorem. If F and G are both antiderivatives of f on an interval, then $F - G$ is constant. So the family $F + C$ is **all** of them.



every member has the same derivative — only C differs

*One shape, shifted. At any x the tangents are parallel
— that is what equal derivatives means.*

READ THE PICTURE

Every member of the family has the same gradient at each x . Knowing the derivative fixes the **shape** of the curve but not its **height**. One extra fact — a point on the curve — fixes the height.

The notation

$$\int f(x) dx = F(x) + C \text{ where } F'(x) = f(x)$$

Read “the integral of f of x , dee x ”. The $\int$ sign is an elongated S, for *summa*; the dx says which letter is the variable. The $+ C$ is part of the answer, not decoration.

+ C IS WORTH A MARK, EVERY TIME

An indefinite integral without its constant is an incomplete answer. It is the single most frequently lost mark in the whole of integration.

The particular antiderivative

An initial condition picks one member of the family:

Example. Find F with $F'(x) = 3x^2$ and $F(2) = 11$.

$$F(x) = x^3 + C \Rightarrow 8 + C = 11 \Rightarrow C = 3 \Rightarrow F(x) = x^3 + 3$$

In physics this is how a motion is reconstructed: the acceleration gives the velocity up to a constant, which the initial velocity supplies; the velocity gives the displacement up to another constant, which the initial position supplies.

$$a(t) = -10 \Rightarrow v(t) = -10t + v_0 \Rightarrow s(t) = -5t^2 + v_0t + s_0$$

Every formula of Grade-9 kinematics is an antiderivative, taken twice.

02 Worked examples

EXAMPLE 1 Find $\int 4x^3 dx$.

① What differentiates to $4x^3$?

② $(x^4)' = 4x^3$

③ Add the constant.

ANSWER

$$x^4 + C$$

EXAMPLE 2 Find the antiderivative of $f(x) = 6x^2 - 4x$ passing through $(1, 5)$.

① $F(x) = 2x^3 - 2x^2 + C$

Check: $F' = 6x^2 - 4x$ ✓

② $F(1) = 2 - 2 + C = C$

③ $C = 5$

ANSWER

$$F(x) = 2x^3 - 2x^2 + 5$$

EXAMPLE 3 A stone is dropped from 45 m. With $a = -10 \text{ m/s}^2$, when does it land?

① $v = -10t + 0$
Dropped, so $v_0 = 0$.

② $s = -5t^2 + 45$
 $s_0 = 45$.

③ $-5t^2 + 45 = 0 \Rightarrow t^2 = 9$

④ $t = 3 \text{ s}$.

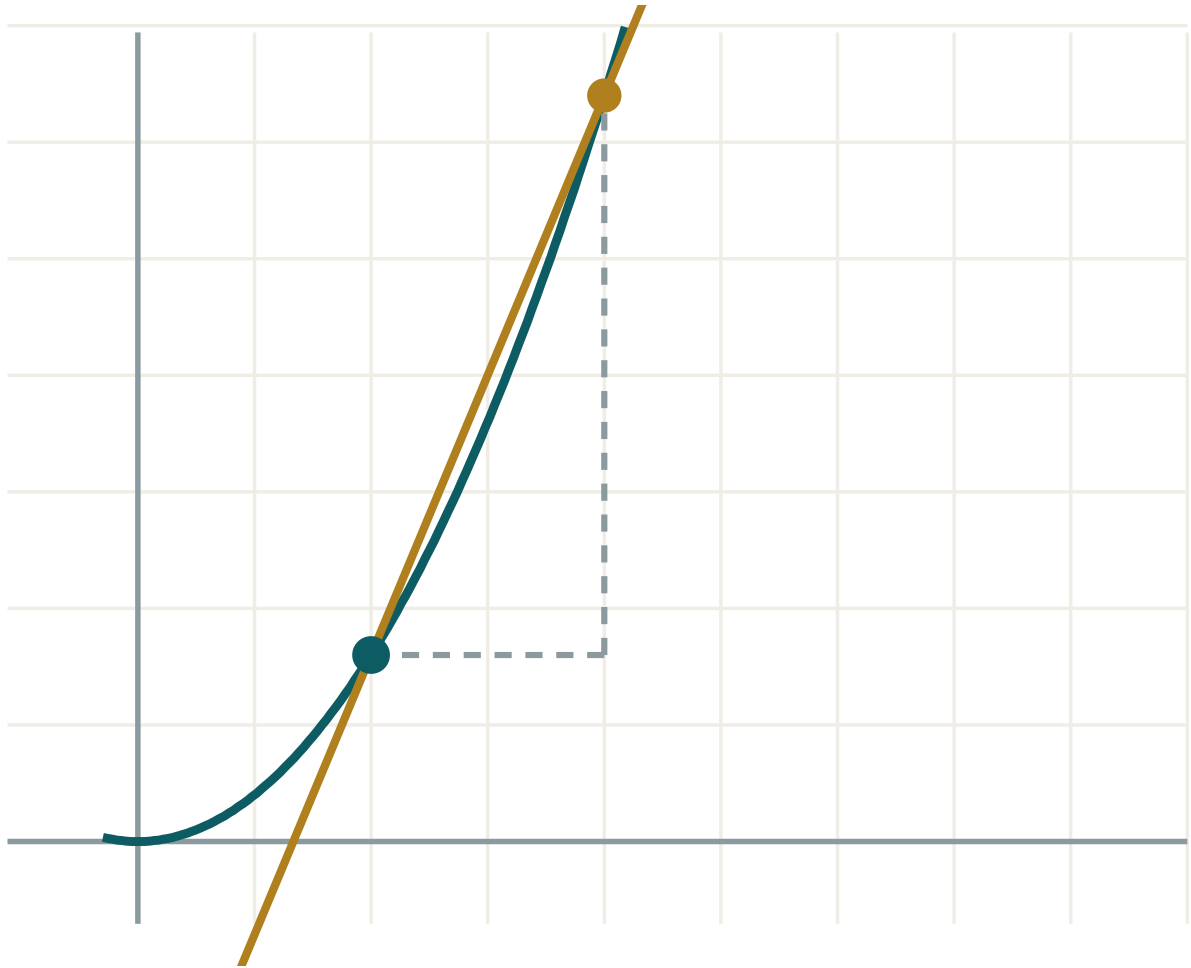
ANSWER

$$t = 3 \text{ s}$$

03 Interactive model

Differentiate every answer on the board before moving on.

Forwards and backwards



x 2

h 2

 $\Delta y / \Delta x$ 2.4 $f'(x)$ exactly 1.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Antiderivative	Boshlang'ich funksiya	Первообразная
Indefinite integral	Aniqmas integral	Неопределённый интеграл
Constant of integration	Integrallash doimiysi	Постоянная интегрирования
Integrand	Integral ostidagi funksiya	Подынтегральная функция
Family of curves	Egri chiziqlar oilasi	Семейство кривых
Initial condition	Boshlang'ich shart	Начальное условие
Particular solution	Xususiy yechim	Частное решение
Integration	Integrallash	Интегрирование
Inverse operation	Teskari amal	Обратная операция

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

F is an antiderivative of f when:

$$F = f$$

$$F' = f$$

$$f' = F$$

$$F + f = 0$$

$\int 2x \, dx$ is:

$$x^2$$

$$x^2 + C$$

$$2$$

$$2x^2 + C$$

Two antiderivatives of the same function differ by:

nothing

a constant

a linear term

anything

$\int 0 \, dx$ is:

To fix C you need:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\int 2x \, dx$

ANSWER $x^2 + C$

2. $\int 3x^2 \, dx$

ANSWER $x^3 + C$

3. $\int 1 \, dx$

ANSWER $x + C$

4. $\int 5 \, dx$

ANSWER $5x + C$

5. $\int 4x^3 \, dx$

ANSWER $x^4 + C$

6. $\int 0 \, dx$

ANSWER C

7. Is $x^2 + 7$ an antiderivative of $2x$?

ANSWER yes

Medium · the standard question

7 PROBLEMS

1. $\int (6x^2 - 4x) dx$

ANSWER $2x^3 - 2x^2 + C$

2. $\int (x + 3) dx$

ANSWER $\frac{x^2}{2} + 3x + C$

3. $\int (2x^3 - x) dx$

ANSWER $\frac{x^4}{2} - \frac{x^2}{2} + C$

4. Antiderivative of $6x^2 - 4x$ through $(1, 5)$

ANSWER $2x^3 - 2x^2 + 5$

5. Antiderivative of $4x$ through $(2, 10)$

ANSWER $2x^2 + 2$

6. $a = -10, v_0 = 0, s_0 = 45$. Find $s(t)$

ANSWER $45 - 5t^2$

7. When does that stone land?

ANSWER $t = 3 \text{ s}$

Hard · stretch and reason

7 PROBLEMS

1. Find F with $F''(x) = 6x$, $F'(0) = 2$ and $F(0) = 1$

ANSWER $F(x) = x^3 + 2x + 1$

2. Antiderivative of $(2x + 1)^2$ — expand first

ANSWER $\frac{4x^3}{3} + 2x^2 + x + C$

3. A car decelerates at 4 m/s^2 from 20 m/s . How far before it stops?

ANSWER 50 m

4. Prove that two antiderivatives differ by a constant

ANSWER $(F - G)' = 0$ on an interval

5. Show no f has $f'(x) = 3x^2$ and $f(1) = f(-1)$

ANSWER $f = x^3 + C$ gives $1 + C = -1 + C$, which is false

6. Why must the interval be stated in the theorem?

ANSWER On a disconnected domain the difference can jump

7. $\int (x + \frac{1}{x})^2 dx$ — expand and integrate the powers

ANSWER $\frac{x^3}{3} + 2x - \frac{1}{x} + C$

07 Homework

Homework — 6 tasks

Differentiate every answer to check it, and never omit $+ C$.

- Find $\int 5x^4 dx$, $\int (3x^2 + 2x) dx$ and $\int (x - 4) dx$.
- Find the antiderivative of $8x^3 - 6x$ passing through $(1, 0)$.
- Find F with $F'(x) = 12x^2$ and $F(2) = 40$.
- A ball is thrown up at 25 m/s from ground level, with $a = -10$. Find $s(t)$ and the greatest height.

5. Explain in three sentences why the answer to an indefinite integral is a family of curves.
6. Show that $x^2 + 3$ and $x^2 - 8$ are both antiderivatives of $2x$, and state their difference.